Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Acid is extremely dangerous and can damage the machinery and can also cause serious burns. Always provide a tank of strong soda water as a neutralizing agent when servicing the batteries. Wear goggles and protective clothing to reduce the possibility of serious personal injury.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

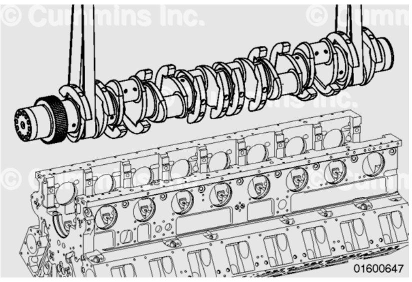
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating engine oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Remove the full flow filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html)
- Remove the engine and place on an engine stand. Refer to Procedure 000-001 in Section 0. (/qs3/pubsys2/xml/en/procedures/56/56-000-001.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)
- Remove the oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)
- Remove the flexplate, if installed. Refer to Procedure 016-004 in Section 16. (/qs3/pubsys2/xml/en/procedures/56/56-016-004-tr.html)
- Remove the flywheel, if installed. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/56/56-016-005-tr.html)

- Remove the flywheel housing. Refer to Procedure 016-006 in Section 16. (/qs3/pubsys2/xml/en/procedures/56/56-016-006-tr.html)
- Remove the front gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html)
- Remove the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html)
- Remove the main bearings caps. Refer to Procedure 001-006 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-006.html)
- Remove the thrust bearings. Refer to Procedure 001-007 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-007-tr.html)

Remove

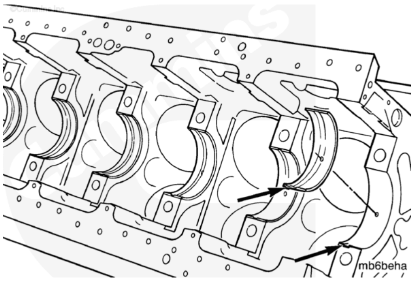
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use a nylon sling and a hoist to remove the crankshaft.

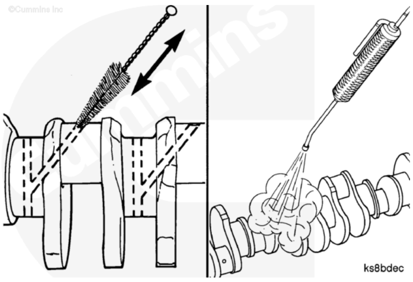
Remove and mark the upper main bearings. Mark the location of the bearing on the tang.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the crankshaft with solvent and dry with compressed air.

Inspect the crankshaft for nicks or scratches. If a mark can **not** be felt with the fingernail, there is no value in polishing the crankshaft.

Use a Scotch-Brite™ 7448 abrasive pad, Part Number 3823258, or equivalent, to remove discoloration or light scratches from the machined surfaces.

The crankshaft **must** be cleaned again if this polishing is performed.

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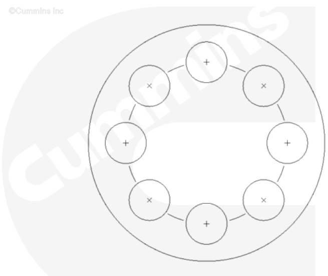
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not use a thread chaser to clean the capscrew threads in the crankshaft. Severe engine damage can result.

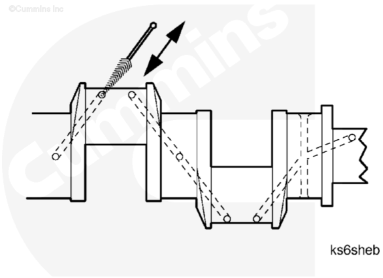
Both the QSK45 and the QSK60 engines have rolled capscrew threads in the nose of the crankshaft.

Use solvent to clean the rolled capscrew threads and dry with compressed air.

Place tape over the threaded capscrew holes.

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

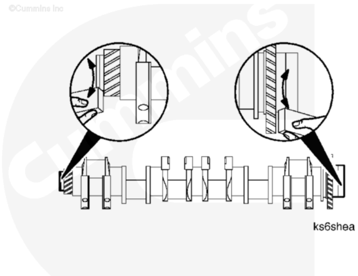


Use a bristle brush and solvent to clean all the crankshaft oil drillings.

Use a light preservative oil to lubricate the crankshaft to prevent the formation of rust.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

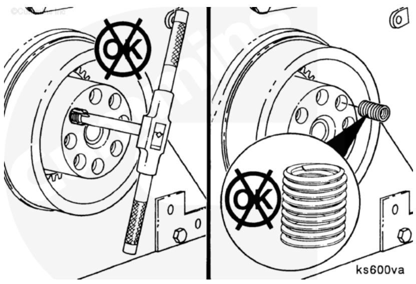


Note : New crankshafts are coated with a heavy preservative. Use solvent to thoroughly remove the coating. Brush or flush the packing debris from the oil drillings before installing the crankshaft into the engine.

Use a honing stone to polish the outside diameter of the front and rear oil seal locations, the flywheel mounting location, and the vibration damper location. Remove all small scratches and grooves.

⚠ CAUTION ⚠

The QSK45 and QSK60 engines use rolled threads in the crankshaft nose. Do not use a thread cutting tap to chase or clean the threads. Damage or failure will result.



Thread salvage inserts are **not** permitted in the crankshaft nose.

Check the threads in the capscrew holes for damage at both ends of the crankshaft.

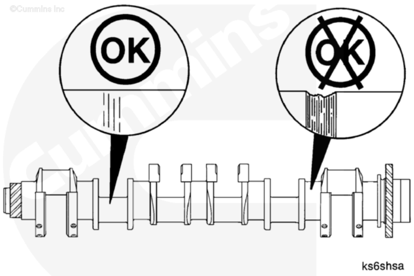
Check the roll pins for damage. Replace the pins if they are damaged.

The roll pins **must** be installed.

The roll pins align the vibration damper and the flywheel with the crankshaft.

The vibration damper and the flywheel contain the index marks required for proper adjustment of the valves and injectors.

Check the main bearing journals and the rod bearing journals for damage or excessive wear. Minor scratches are acceptable.



If a fingernail is caught in the scratch, the crankshaft **must** be replaced or machined. Reference the Alternative Repair Manual, Bulletin 3379035.

Measure the connecting rod and main bearing crankshaft journal outside diameter.

Crankshaft Main Bearing Journal Outside Diameter			
	mm		in
Standard	184.10	MIN	7.248
	184.15	MAX	7.250
Undersize 0.25 mm [0.010 in]	183.85	MIN	7.238
	183.90	MAX	7.240
Undersize 0.50 mm [0.020 in]	183.60	MIN	7.228
	183.65	MAX	7.230
Undersize 0.75 mm [0.030 in]	183.35	MIN	7.218
	183.40	MAX	7.220
Undersize 1.016 mm [0.040 in]	183.10	MIN	7.208
	183.15	MAX	7.210

Crankshaft Connecting Rod Journal Outside Diameter			
	mm		in
Standard	120.57	MIN	4.747
	120.65	MAX	4.750

Undersize 0.25 mm [0.010 in]	120.32	MIN	4.737
	120.40	MAX	4.740
Undersize 0.50 mm [0.020 in]	120.07	MIN	4.727
	120.15	MAX	4.730
Undersize 0.75 mm [0.030 in]	119.82	MIN	4.717
	119.90	MAX	4.720
Undersize 1.016 mm [0.040 in]	119.57	MIN	4.707
	119.65	MAX	4.710

The crankshaft can be machined undersize if the outside diameter is **not** within specifications. **Always** grind all of the journals when one is **not** within specifications.

Oversize rod bearings and main bearings are available. Reference the Alternative Repair Manual, Bulletin 3379035, for grinding specifications and instructions.

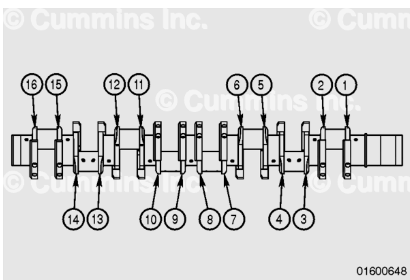
Oversize thrust bearings are available if the thrust distance is **not** within specifications. Bearings that are 0.25 mm [0.010 in], 0.51 mm [0.020 in], 0.76 mm [0.030 in], and 1.016 mm [0.040 in] thicker than standard are available. If the crankshaft **must** be machined for an oversize thrust bearing, refer to the Alternative Repair Manual, Bulletin 3379035.

Use a light preservative oil to lubricate the crankshaft to prevent the formation of rust. If the crankshaft is **not** going to be installed immediately, use a heavy preservative oil.

Note : The illustration shows the QSK60 crankshaft, which contains 16 webs. The QSK45 crankshaft contains 12 webs.

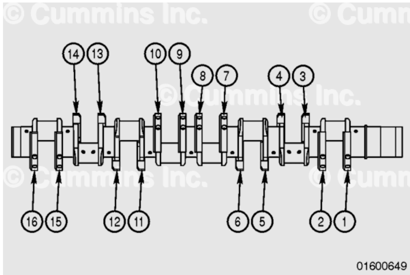
The webs of the crankshaft are numbered from the front of the crankshaft. The number is steel stamped on the web.

In the illustration, the numbers (1) through (16) indicate the location of the stamps.



⚠ CAUTION ⚠

If the counterweights are removed, they must be installed in the correct position on the crankshaft from which they were removed. If they are installed in the wrong position, the crankshaft will fail.

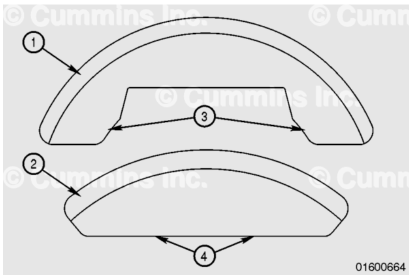


The QSK45 and QSK60 crankshafts contain counterweights that are attached with capscrews. The counterweights have the same number stamped on them as the web to which they are attached, as illustrated (1) through (16).

If the counterweight is replaced, the crankshaft **must** be balanced. Reference the Alternative Repair Manual, Bulletin 3379035.

⚠ CAUTION ⚠

Steel and tungsten counterweights must never be mixed on a crankshaft or engine damage can occur.

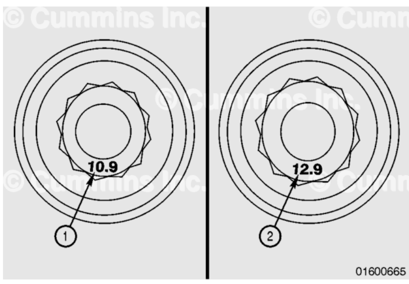


Note : The QSK60 has both fully and partially counterweighted crankshafts. The fully counterweighted crankshaft has sixteen counterweights with one on each crankshaft web. The partially counterweighted crankshaft has eight counterweights.

QSK60 engines use steel (1) counterweights, while QSK45 engines may use steel (1) or tungsten (2) counterweights. The steel and tungsten counterweights can be identified by their difference in shape. The steel counterweight has extensions (3) while the tungsten (2) counterweight has a flat base (4).

⚠ CAUTION ⚠

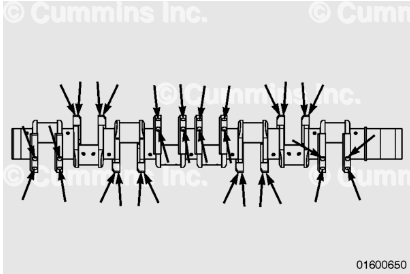
Make sure the tungsten counterweight is installed using the 12.9 capscrew and the appropriate torque value. If the incorrect capscrew or torque value is used, engine damage can occur.



Due to the difference in weight between the two counterweight materials, different counterweight capscrews and capscrew torque values are used. The correct counterweight capscrew can be identified by the steel properties marking on the top of the capscrew head. The steel counterweight capscrew is marked with the number 10.9 (1) while the tungsten counterweight capscrew is marked 12.9 (2).

⚠ CAUTION ⚠

Note any loose capscrews. The capscrews must be tight or the counterweight can fall, causing severe damage to the engine. Any loose capscrews must be replaced with new capscrews.



⚠ CAUTION ⚠

Capscrews can not be used again and must be replaced with new capscrews.

⚠ CAUTION ⚠

Check new capscrews for any visible damage or wear. Discard the capscrew if there is any visible damage or wear to the thread, neck, or capscrew head.

Each counterweight contains two capscrews.

Use a torque wrench to check the torque of the counterweight capscrews.

Check which capscrew is installed. Use a torque wrench to check the torque of the counterweight capscrews.

Torque Value:
10.9 Capscrews 240 n•m [177 ft-lb]

Torque Value:
12.9 Capscrews

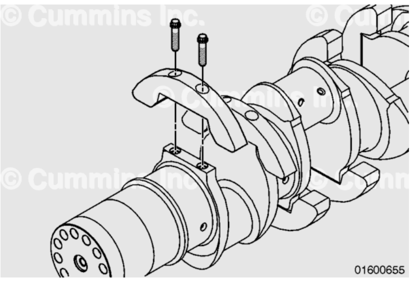
1. 40 n•m [30 ft-lb]
2. Loosen
3. 40 n•m [30 ft-lb]
4. Turn 90 degrees

Mark any counterweights that have a loose capscrew.

Note : The following steps apply only to the counterweights that contain a loose capscrew.

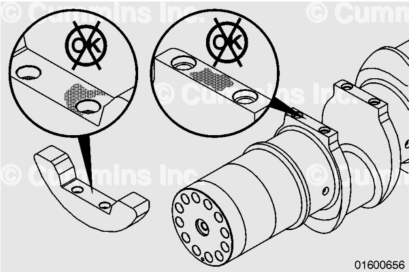
Remove the capscrews and discard.

Remove the counterweight.



⚠ CAUTION ⚠

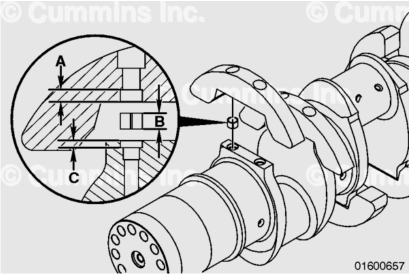
Fully counterweighted crankshafts can not be repaired at the number 14 and 15 counterweight locations. If fretting occurs in these locations, the crankshaft must be replaced. Repair of crankshafts at these locations may result in engine damage.



Check the counterweight and crankshaft for fretting or damage between the mating surfaces. If the counterweight is damaged, the counterweight **must** be replaced and the crankshaft **must** be balanced. If the crankshaft is damaged, the crankshaft **must** be replaced or the counterweight replaced and the crankshaft repaired.

Not more than 0.51 mm [0.020 in] can be removed from the crankshaft mounting pad to repair fretting.

Before installing the counterweight to the crankshaft, make sure there is sufficient clearance between the crankshaft, counterweight, and dowel.



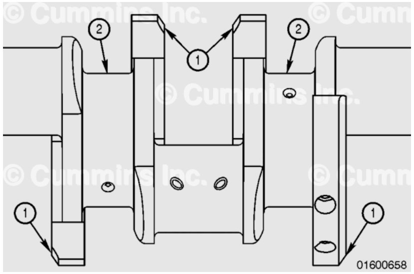
Measure the depth of the dowel bore in both the counterweight (A) and the crankshaft (C). Measure the height of the dowel (B).

The total depth of both the dowel bore in the counterweight (A) and the crankshaft (C), (A+C), **must** be greater than the height of the dowel (B). If the dowel height (B) is greater than this total depth (A+C) the counterweight will **not** sit correctly on the crankshaft and engine damage will occur.

If the dowel height (B) is greater than this total depth (A+C) a maximum of 0.51mm [0.020 in] of material can be removed from one end of the dowel. Remove any burrs created as a result of material removal.

⚠ CAUTION ⚠

If the counterweight is removed, it must be installed with the chamfered edge (1) facing away from the main bearing journal (2). If the counterweight is not installed correctly, the weight can strike the connecting rod, causing extensive damage to the engine.



⚠ CAUTION ⚠

Make sure the correct capscrew and torque value is used for the type counterweight installed.

Note : Make sure the counterweight is returned to its original location. The counterweights have the same number stamped on them as the web to which they are attached.

Lubricate the capscrew head and threads with clean oil.

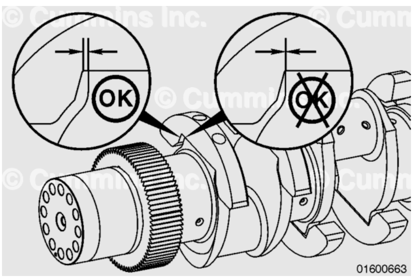
Install the capscrews.

Torque Value:
10.9 Capscrews 240 n•m [177 ft-lb]

Torque Value:
12.9 Capscrews

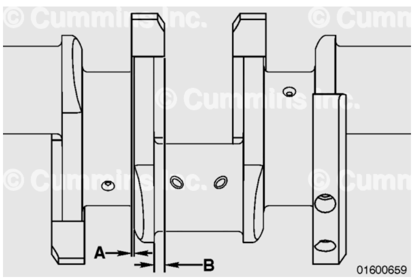
1. 40 n•m [30 ft-lb]
2. Loosen
3. 40 n•m [30 ft-lb]
4. Turn 90 degrees

With the capscrews tightened, inspect the counterweight for an interference condition with the crank. There **must not** be an interference between the crankshaft and counterweight to the sides of the mounting pad.



Check the overhang of the counterweight to the main and connecting rod bearing journal side walls. The overhang **must not** exceed the maximum limits called out in the following table.

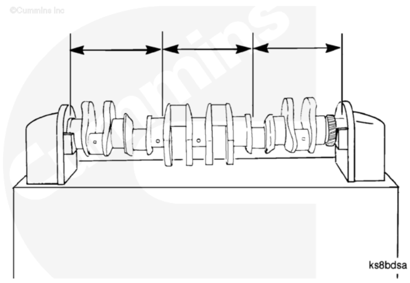
If the overhang exceeds either of these maximum limits, the counterweight **must** be replaced. If the overhang exceeds either of these maximum limits with a new counterweight, the crankshaft **must** be replaced.



Note : If a new counterweight is fitted, the crankshaft must be rebalanced.

Engine	Dimension	Counterweight/ Web Location	Overhang	
mm	inch			
QSK45	A	All locations	4	0.157
QSK45	B	All locations	15	0.591
QSK60	A	1-13 and 16	4	0.157
QSK60	A	14 and 15	3	0.118
QSK60	B	All locations	15	0.591

Magnetic Crack Inspect



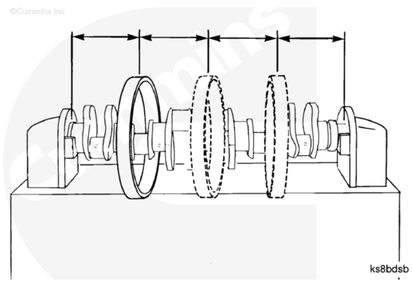
Use a magnetic particle testing machine for the test.

Perform the head shot and inspection procedure, then perform the coil shot and inspection procedure.

Adjust the testing machine to 1800 amperes direct current or rectified alternating current.

Use the continuous method. Wet **only** ⅓ of the crankshaft at a time.

Check the crankshaft for cracks.



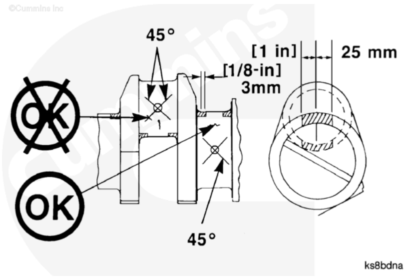
Use the coil shot method. Use a coil that is a minimum of 514.0 mm [20.250 in] diameter.

Use the continuous method. Apply a coil shot.

Amperage (Ampere Turns)	
Minimum	**Maximum**
4500 direct current or rectified alternating current	5000 direct current or rectified alternating current

Ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

An open indication is visible after the wetting operation has been completed. An indication below the surface is **not** visible after the wetting operation has been completed. An indication below the surface can be seen with the use of the ultraviolet light.

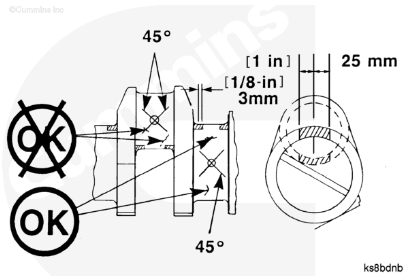


Do **not** use the crankshaft if:

Limits of Acceptance - Open Indications

- There is an indication in the fillet or in the shaded area.
- There is an indication that passes through the 45 degree diagonal from the oil hole or goes into the oil hole chamfer.
- There is an indication that is longer than 3.0 mm [0.12 in] and must **not** enter the oil hole.
- Total cumulative length of open longitudinal indications on any one big end or main journal does **not** exceed 15.0 mm [0.601 in].
- Circumferential indications of any kind are **not** acceptable.

Do **not** use the crankshaft if:

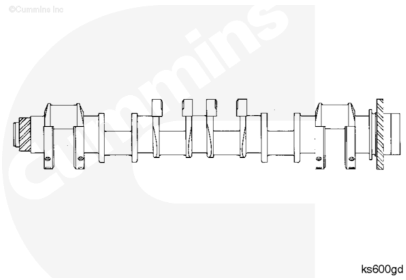


Limits of Acceptance - Indications Below the Surface

- There is an indication in the fillet or in the shaded area that is in a circumferential direction.
- There is an indication in a longitudinal direction that is longer than 3.0 mm [0.12 in].
- There is an indication that is closer than 1.5 mm [0.063 in] to an oil hole chamfer.
- There is an indication that passes through the 45 degree diagonal from the oil hole.
- Circumferential indications of any kind are **not** acceptable.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ CAUTION ⚠

The crankshaft must be demagnetized completely and cleaned thoroughly. Small metal particles will cause engine damage.

Use steam to clean the crankshaft and oil drillings.

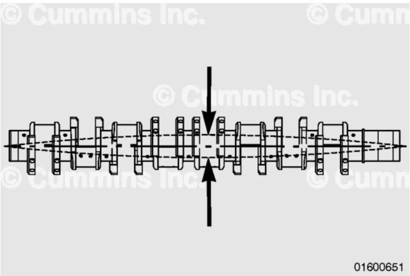
Use a light preservative oil to lubricate the crankshaft to prevent the formation of rust.

Use a heavy preservative oil if the crankshaft is **not** going to be installed immediately. Protect the part with a cover to prevent dirt from sticking to the oil.

Bend and Twist Inspect

The crankshaft straightness is determined by the amount of total indicator run out and adjacent journal runout.

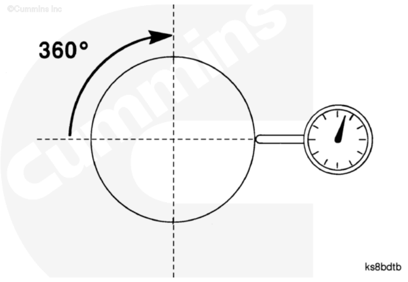
Total indicator runout is measured at the middle bearing journal when the crankshaft is supported on the two end journals.



Journal runout is defined as the total indicator runout (total sweep of the needle) of the main bearing journals when the crankshaft is rotated one complete revolution (360 degrees).

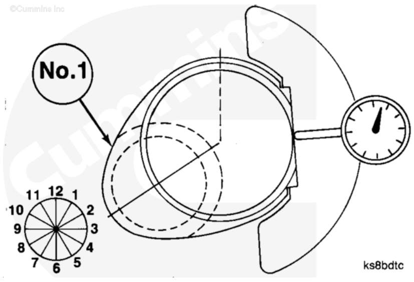
Adjacent journal runout is defined as the relationship of the total indicator runout of a main bearing journal as it is rotated on a common axis to the total indicator runout of an adjacent journal.

Adjacent runout is often referred to as step runout, bearing-to-bearing runout, or journal-to-journal runout.



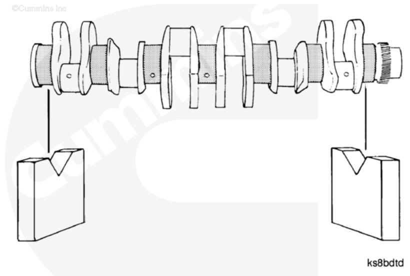
The clock position is defined as the location of the journal at the highest total indicator runout point. Compare its angular relationship with the number 1 crankshaft pin as viewed from the front of the crankshaft.

In the illustration, the crankshaft pin is at the 8 o'clock position. This is the clock position of the journal being measured.



⚠ WARNING ⚠

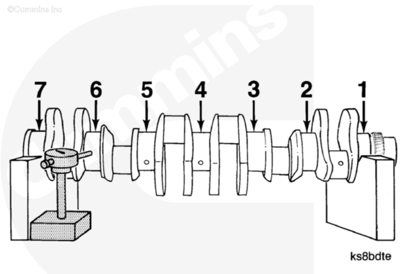
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Place two V-blocks on a flat surface.

Support the crankshaft on the V-blocks at the two end bearing journals.

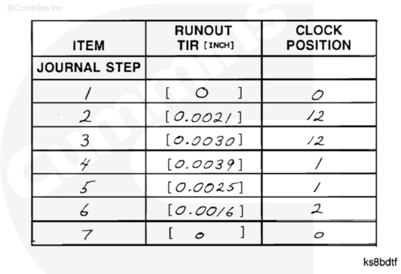
Note : The QSK45 crankshaft is shown in the illustration. The QSK60 has nine main bearing journals.



Position a dial indicator so the stem touches the center line of the main bearing journal.

The dial indicator **must** be positioned at the centerline of any journal that is measured, (1) through (7).

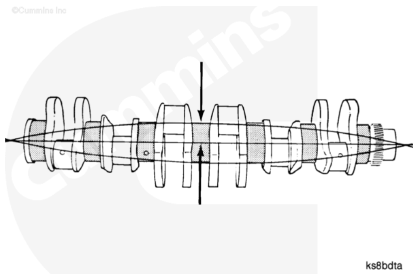
Rotate the crankshaft and measure the total indicator runout at each bearing journal. Record the value and the clock position for each location.



ITEM	RUNOUT TIR [INCH]	CLOCK POSITION
JOURNAL STEP		
1	[0]	0
2	[0.0021]	12
3	[0.0030]	12
4	[0.0039]	1
5	[0.0025]	1
6	[0.0016]	2
7	[0]	0

ks8bdtf

All QSK45 and QSK60 crankshafts are fully fillet hardened.



A fully fillet hardened crankshaft **must** be discarded if the bow is **not** within specifications.

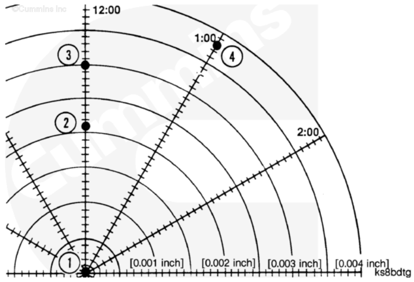
Position the dial indicator stem to the center main bearing journal.

Rotate the crankshaft a complete revolution.

Crankshaft Run-out		
mm		in
0.230	MAX	0.009

If the crankshaft is **not** within specifications, it **must** be replaced.

For each journal, plot the total indicator runout value at its clock position on a polar chart, as illustrated on the next page.



The end journals, supported by V-blocks, **must** be plotted at the center of the chart.

The graphic illustrates the plot points.

Journal	Total Indicator Runout	Clock Position
(1)	0	0
(2)	0.002	12
(3)	0.003	12
(4)	0.004	1

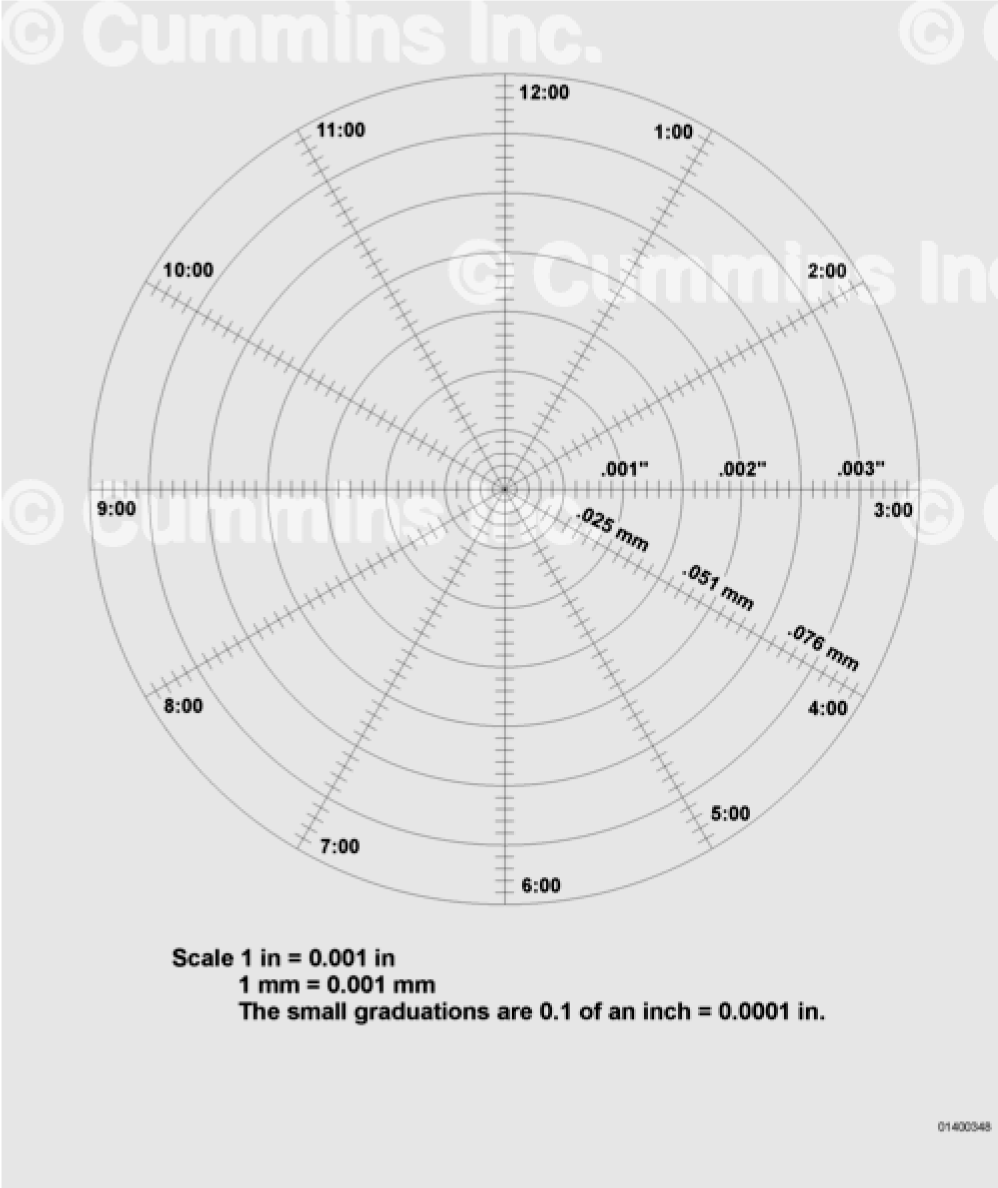
Draw a straight line between the plotted points. From journal number 1 to journal number 2, from journal number 2 to journal number 3, until all journals are plotted on the chart.

To determine the adjacent journal runout, measure the length of the line from each journal to its corresponding journal point.

Record the adjacent journal runout for each main bearing journal.

Adjacent Journal Run-out		
mm		in
0.08	MAX	0.003

If the crankshaft is **not** within specifications, it **must** be replaced.



Polar Chart

Scale 1 inch = 0.001 inch and 1 mm = 0.001 mm

The small graduations are 0.1 of an inch = 0.0001 in.

A fixed central support should **never** be used to remove crankshaft bending or sag. A fixed central support will over-constrain the crankshaft and impart errors to the measured data as the crankshaft turns.

A central support that is free to float in the direction of the measurement may be used to mitigate crankshaft sag (deflection of the component under its own weight). If a floating central support is used, the support **must** be free to move with very little resistance in the direction perpendicular to the crankshaft, even when supporting the weight of the crankshaft.

The use of a floating central support requires a correlation study to validate its impact on crankshaft measurement. This study involves measuring total indicator runout using two setups: first, with the crankshaft supported **only** on the endmost journals; and second, with the crankshaft supported on the endmost journals as well as on the central journal using the floating central support. The study **must** demonstrate that the difference in total indicator runout between the two setups does **not** exceed 15 percent of the crankshaft's runout specification.

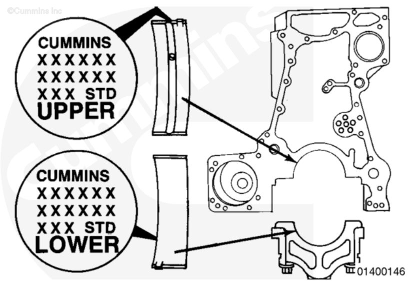
Engine Configuration	Support Location		
V16*	Front	Center	Rear
	1	5	9
*A floating central support may **only** be used if measuring in the horizontal direction, and subject to the requirements above. A fixed central support should **not** be used.			

Install

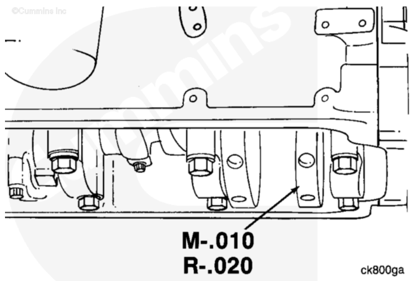
The upper bearings contain an oil hole. The lower bearings do **not** have an oil hole.

Both bearings are marked on the back side indicating location (UPPER or LOWER) and the size (STANDARD [STD] or OVERSIZE [OS]). If the bearing is oversize, the amount of oversize is indicated in U.S. customary inches.

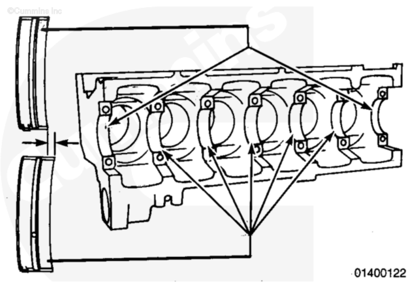
If the crankshaft was **not** ground, use the same size bearing that was removed [STD, 0.010, 0.020, or 0.030].



The crankshaft will be stamped on the end of the number 1 counterweight to indicate if it has been ground under size. Thrust bearing size is stamped on a crankshaft counterweight adjacent to the thrust location.

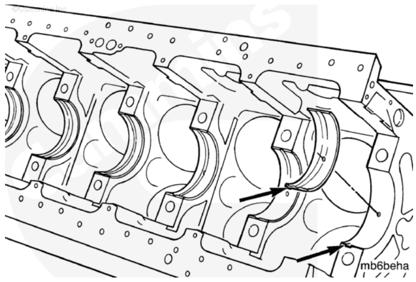


The location of the main bearing caps, beginning from the front of the engine block, are number 1 through number 7 for the QSK45, and number 1 through number 9 for the QSK60.



⚠ CAUTION ⚠

Prevent dirt from mixing with the lubricant. Dirty lubricant can cause low hour failures.



Use a lint-free cloth to clean the main bearings and main bearing mounting surfaces.

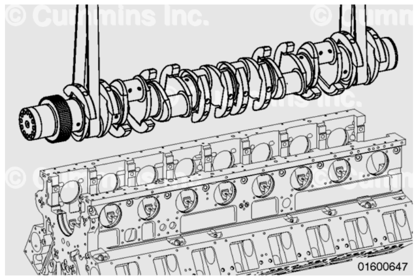
Do **not** lubricate the backs of the main bearings.

Align the tang in the main bearing with the slot in the engine block. Install the bearings. The end of the bearing **must** be even with the mounting surface for the main bearing cap.

Lubricate the bearings with Lubriplate™ 105, or equivalent.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use a lint-free cloth to clean the crankshaft bearing journals.

Install the crankshaft with the gear at the front of the engine block.

Note : The QSK45 tungsten counterweighted crankshaft **must only** be installed in engines intended for use in certain applications. Make sure the intended engine application is appropriate for use with this crankshaft.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Acid is extremely dangerous and can damage the machinery and can also cause serious burns. Always provide a tank of strong soda water as a neutralizing agent when servicing the batteries. Wear goggles and protective clothing to reduce the possibility of serious personal injury.

⚠ **WARNING** ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the thrust bearings. Refer to Procedure 001-007 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-007-tr.html)
- Install the main bearings caps. Refer to Procedure 001-006 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-006.html)
- Install the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html)
- Install the front gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html)
- Install the flywheel housing. Refer to Procedure 016-006 in Section 16. (/qs3/pubsys2/xml/en/procedures/56/56-016-006-tr.html)
- Install the flywheel, if removed. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/56/56-016-005-tr.html)
- Install the flexplate, if removed. Refer to Procedure 016-004 in Section 16. (/qs3/pubsys2/xml/en/procedures/56/56-016-004-tr.html)
- Remove the engine from the engine stand and install in the chassis. Refer to Procedure 000-002 in Section 0. (/qs3/pubsys2/xml/en/procedures/56/56-000-002.html)
- Install the oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)
- Install the full flow filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for leaks.

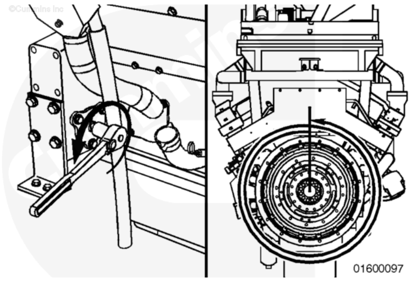
Rotation Check

To rotate the engine crankshaft, push in on the engine barring device and rotate **counterclockwise**.

Rotate the crankshaft two complete revolutions.

If the engine does **not** turn freely, the equipment can have a malfunction. See equipment manufacturer service information.

The engine can have internal problems. See the correct procedure for inspection and replacement of internal engine components.



Last Modified: 29-Jul-2025